
The Pro-Action Operator: A Scientifically-Informed, Multi-Timescale Self-Regulation Model for LLM-Agents

Anonymous Author(s)

Affiliation

Address

email

Abstract

Contemporary LLM-agent frameworks — ReAct, LangGraph, OpenClaw — have prioritized *capability* over *motivation*, deciding *when* an agent acts through exogenous flow control or task-backlog heartbeats. Homeostatic reinforcement learning (Keramati–Gutkin 2014) and active-inference agents (Friston 2017; Tschantz et al. 2020) internalize regulation, but often through a compact objective rather than an explicit set of delayed regulatory subsystems. However, biological regulation is not single-timescale: converging work in stress physiology, self-regulation, affect, and cognitive control treats adaptation as the interaction of subsystems — attention, perception, hormonal (SAM adrenergic, seconds; HPA cortisol, minutes), emotional, neuropsychological, cognitive — exchanging signals across distinct delays, consistent with the mechanism behind Yerkes–Dodson inversion and delayed reappraisal (Porcelli–Delgado 2017). We propose the **Pro-Action operator** Γ , a recursively composable, multi-timescale thermostat system. Each subsystem k updates as

$$x_{k,t+1} = x_{k,t} - \kappa_k(x_{k,t} - x_{k,t}^*) + \lambda_k - \alpha_k \rho_k(a_t, e_{t+1}) + [W \phi(\mathbf{x}_{t-\tau_k})]_k,$$

with $\phi = \tanh$ elementwise, subsystem-specific delay τ_k encoding a dimensionless SAM–HPA-like ratio ($\tau_k^{\text{HPA}} \gg \tau_k^{\text{A}}$), and action selected by an arousal-gated mixer. Γ offers a finite, inspectable parameterization of regulation: the τ_k are explicit hyperparameters rather than hidden in orchestration depth. We specify three falsifiable properties a priori: (P1) elastic return to set-point; (P2) Yerkes–Dodson inverted-U; (P3) delayed reappraisal. As a Concept & Feasibility contribution, the paper reports the model, a lightweight falsifiability protocol, computational sanity checks, and a completed LLM-agent benchmark. The completed benchmark shows that the operator produces opponent-discriminative behavior that exogenous-activation baselines (ReAct) and structurally similar reductions (Random- Γ , Collapse-NC) do not — a feasibility claim, not a causal claim about the delay structure specifically.

Keywords. Homeostatic reinforcement learning; LLM agents; active inference; multi-timescale self-regulation; endogenous activation; cybernetics; self-regulation model.

1 Introduction

The deployment of LLM-based autonomous agents into settings of institutional consequence — health triage, financial advice, public deliberation — is moving faster than the science of how such agents should *regulate themselves*. Whether an agent speaks, waits, defers, or escalates tends to be decided outside the agent, by an orchestrator whose control rules may be less expressive than the situations they coordinate: a small set of graph transitions, queues, or heartbeats is used to govern

high-dimensional and shifting social or task states. In Ashby’s terms, the controller may lack the requisite variety needed to match the variety of the environment [1].

A frequently reported symptom, consistent with this variety mismatch, is the loss of activation coherence in adversarial multi-agent deliberation: floor monopolization, context collapse, and interventions that appear decoupled from any internal state the agent could be said to carry (evidence observed by the authors in a companion work, currently under review). The pattern suggests that *when* an agent acts may depend on structure the dominant paradigm does not explicitly represent.

Contemporary LLM-agent frameworks — ReAct, LangGraph, OpenClaw — have prioritized *capability* (what an agent can do) over *motivation* (why and when it should act at all), addressing activation via cognitive while-loops and exogenous flow control. Homeostatic reinforcement learning (Keramati–Gutkin 2014) and active-inference agents (Friston 2017; Tschantz et al. 2020) internalize regulation, but do not provide a compact LLM-agent operator in which attention, interoception, emotion, memory, cognition, and stress-like physiology are all explicit delayed regulators. Intelligent animals, including humans, regulate behavior through multiple interacting systems — attention, arousal, affect, memory, deliberation — operating on distinct timescales whose relative priority shifts with context; dominant LLM-agent paradigms treat activation as a single-objective signal or exogenous flow control.

The problem is older than LLM-agents. Mid-20th century behaviorism defined it precisely by bracketing it: the stimulus–response mapping could be studied scientifically because the intervening machinery — the *black box* — was deliberately left unspecified. Decades of subsequent research opened that box from different angles: behavioral biology established that animal regulation is multi-drive and hierarchical; stress physiology revealed fast (adrenergic, seconds) and slow (cortisol, minutes) adaptation axes with distinct roles; affective neuroscience showed that emotion is not an output label but a construction that shapes perception, memory, and action; evolutionary game theory demonstrated that social behavior is governed by reciprocity and temporal discounting, not instantaneous utility; cybernetics formalized why a regulator must match the variety of what it controls. Each tradition contributed a piece. Nobody has assembled these threads for LLM-agents into a single, compact, falsifiable operator. Section 2 reviews each contribution in detail. Opening the black box matters beyond mechanistic completeness: (i) Γ provides a finite, inspectable state vector, making each subsystem’s contribution auditable rather than buried in prompt engineering; (ii) a controller tracking regulatory state creates the precondition for agents that adapt to their own arousal trajectory; (iii) explicit emotional and hormonal channels provide the computational substrate for empathy modeling — recognizing a counterpart’s affective state — which is structurally distinct from fine-tuning on empathy datasets.

What remains comparatively unexplored is a single, computable, falsifiable operator that integrates these disciplinary insights for LLM agents. The central biological clue is that regulation is not a single controller but a *system of subsystems* — attention, perception, hormonal (SAM adrenergic, seconds; HPA cortisol, minutes), emotional, neuropsychological, cognitive — exchanging signals across distinct delays. The SAM–HPA timescale split is not decorative: it is consistent with the mechanism behind Yerkes–Dodson inversion, post-stress reappraisal windows, and delayed-valuation shifts [2]. The scientific question is therefore whether activation itself can be treated as a regulated internal process — a coupled set of thermostat-like variables whose delayed interactions determine when the agent should speak, wait, defer, or escalate — and whether the smallest such structure remains distinguishable from a single-timescale regulator.

We propose the **Pro-Action operator** Γ as a model, not an experimental technique. Γ treats activation as a recursively composable system of six coupled thermostats with explicit update delays. The contribution type is Concept & Feasibility: the idea is intentionally broader than what can be validated in a single benchmark, but it must still be made operational enough to fail. The contributions of this paper are primarily *modeling*: (i) a formal operator integrating multi-timescale regulation; (ii) the framing of Γ as a Self-Regulation Model (SRM) class complementary to capability-oriented LAMs; (iii) an a priori specified falsifiability protocol; (iv) a reference implementation checked by computational sanity tests; and (v) a preliminary LLM-agent benchmark snapshot used as feasibility evidence rather than final empirical adjudication.

86 2 Related work

87 **LLM-agent orchestration.** Recent LLM-agent frameworks such as ReAct [3], LangGraph, and
88 OpenClaw made language models operational by embedding them in action loops, tool graphs,
89 and multi-agent workflows. This was a necessary engineering step: without explicit structure, an
90 LLM has no durable procedure for acting in an environment. More recently, agent architectures
91 have incorporated persistent memory streams, reflection, and simulated affective states to enrich the
92 content of decisions [4–7]. The limitation common to both generations is that activation is usually
93 externalized: the graph or supervisor decides when the model acts; the model supplies content for
94 that externally selected moment. Γ is complementary: those works provide richer *content*; Γ provides
95 a principled internal account of *timing*.

96 **Internal regulation across timescales.** A different tradition begins from the opposite assumption:
97 action should be driven by an internal regulatory state. Homeostatic RL makes internal deficit part of
98 the reward [8]; active inference gives a broader variational formulation [9, 10]. These approaches
99 are important — they move regulation inside the agent — but do not yield a compact operator
100 with separately inspectable subsystem delays. Once regulation is internalized, the next question is
101 whether a single regulatory signal is sufficient. Stress physiology and allostasis suggest it is not: fast
102 sympathetic responses (SAM, seconds) and slower endocrine/cognitive recovery (HPA, minutes) play
103 distinct adaptive roles [11–13]. The remaining question is which internal variables deserve to be
104 modeled. Interoceptive and somatic-marker accounts connect bodily state to action readiness [14, 15];
105 affect-regulation accounts treat emotion as part of how situations are appraised [16, 17]; evolutionary
106 game theory adds reciprocity and temporal discounting as social constraints [18, 19]. Together, these
107 literatures motivate attention, interoception, emotion, memory, cognition, and hormonal dynamics as
108 functional regulators rather than decorative modules.

109 **Systems, cybernetics, and the gap.** If these variables are treated as separate regulators, the final
110 problem is structural: how should they be composed? Cybernetics supplies the relevant language:
111 Ashby’s law of requisite variety requires a regulator rich enough to match the environment [1]; Beer’s
112 Viable System Model adds recursive organization [20]; systemism frames the relevant object as
113 composition rather than isolated component [21, 22]. The gap might be narrow yet important. To
114 our knowledge, the threads above — exogenous orchestration, homeostatic internalization, multi-
115 timescale physiology, affect regulation, and cybernetic composition — have not been combined into a
116 small, falsifiable LLM-agent operator in which the six regulatory domains are explicit state variables
117 with explicit delays. That is the gap Γ is designed to occupy.

118 3 The Pro-Action operator

119 3.1 Axioms of homeostatic autonomy (A1–A9)

120 Before defining Γ , we make explicit the modeling commitments that the operator assumes. The
121 axioms are not empirical results or theorems; they state, in this model, what counts as an autonomous,
122 self-regulating agent, which internal variables are allowed to regulate action, and how those variables
123 may be composed. This is meant to avoid leaving the agent’s ontology implicit in implementation
124 choices. The axiom set was subjected to a formal coherence check using SMT-based tools (Z3) to
125 verify that no pair of axioms is mutually contradictory under the intended semantics; all checks
126 passed.

127 We extend the Axioms of Homeostatic Autonomy from the companion work (A1–A3) with six
128 additional axioms that make the multi-timescale structure explicit:

129 **A1–A3 (Kernel).** Autonomy, Impulse, and Quality Gate: the agent maintains an internal deficit δ_i
130 that drives action probability $p_i(t) = \sigma(D(\delta_i(t)))$, reduced by action quality g .

131 **A4 (Perception as initial appraisal).** Every stimulus is appraised before selective attention. A
132 perception operator $\mathcal{P}$ produces an initial situational appraisal $\tilde{o}_t = \mathcal{P}(o_t, \mathcal{I}_t, \mathcal{M}_t; x_{\mathcal{P},t})$, encoding
133 what the stimulus means independent of current priorities.

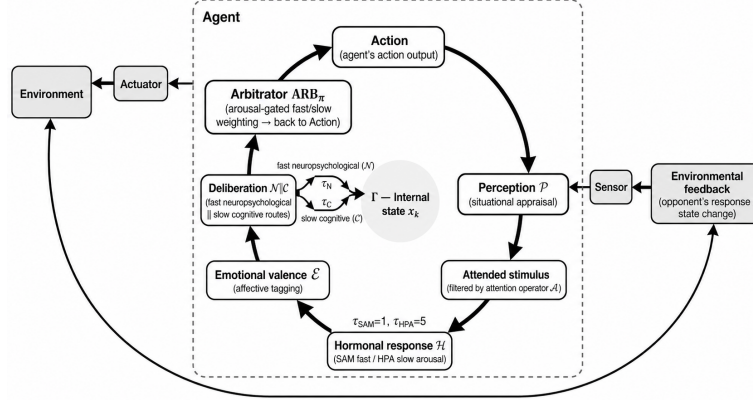


Figure 1: The Pro-Action operator Γ : regulatory cycle in canonical clockwise order. Perception ($\mathcal{P}$, situational appraisal) feeds Attention ($\mathcal{A}$, saliency filtering), which gates the Hormonal arousal module ($\mathcal{H}$, SAM fast / HPA slow), Emotional valence ($\mathcal{E}$), and a Deliberation gate routing to the Neuropsychological fast path ($\mathcal{N}$, τ_N) or Cognitive slow path ($\mathcal{C}$, τ_C); the Arbitrator ARB_π applies arousal-gated weighting before action emission. Notation: $\circ$ denotes composition, $\parallel$ parallel evaluation (routes merged by ARB_π), and $\mathcal{O}[\cdot]$ the output projection selecting the final action.

134 **A5 (Attention as saliency gate).** Not every appraised feature enters the action-selection context. An
 135 attention operator $\mathcal{A}$ filters the perceived signal, selecting $o'_t = \mathcal{A}(\tilde{o}_t, \mathcal{D}_t, \mathcal{M}_t; x_{\mathcal{A},t})$ based on current
 136 drives and memory. Interoceptive signals $\mathcal{I}_t$ feed in parallel to both $\mathcal{P}$ and $\mathcal{A}$.

137 **A6 (Memory as ubiquitous prior).** A state $\mathcal{M}_t$ biases all subsystems and updates post-action.

138 **A7 (Drives as set-point generators).** Set-points are not constants: $\mathbf{x}_t^* = \mathcal{D}_t$.

139 **A8 (Fast/slow parallelism with arbitrator).** Two routes produce action candidates c^{fast} , c^{slow} ; an
 140 arbitrator assigns precision weights π_{fast} , π_{slow} .

141 3.2 The operator

142 We define Γ as the composition (Figure 1):

$$\Gamma := \mathcal{O} \left[\text{ARB}_\pi \left(\underbrace{(\mathcal{N}_{\text{fast}} \circ \mathcal{H}_{\text{SAM}})}_{\text{fast route}} \parallel \underbrace{(\mathcal{C} \circ \mathcal{E} \circ \mathcal{A} \circ \mathcal{P})}_{\text{slow route}} \right) \right] \Big|_{\mathcal{I}, \mathcal{M}, \mathcal{D}} \quad (1)$$

143 Each subsystem k updates as:

$$x_{k,t+1} = x_{k,t} - \kappa_k(x_{k,t} - x_{k,t}^*) + \lambda_k - \alpha_k \rho_k(a_t, e_{t+1}) + [W \phi(\mathbf{x}_{t-\tau_k})]_k \quad (2)$$

144 with $\phi = \tanh$ elementwise. The delays τ_k are dimensionless non-negative integers (in agent-
 145 interaction rounds), not literal biological durations; what the model inherits from SAM-HPA physi-
 146 ology is the *ratio* $\tau_{\mathcal{H}}^{\text{HPA}} \gg \tau_{\mathcal{A}}$, not the absolute timescales. Action is selected by:

$$a_{t+1}^* = \arg \min_a \pi_f \mathcal{F}_{\text{short}} + (1 - \pi_f) \mathcal{F}_{\text{long}} + \gamma \|Q^{1/2}(\mathbf{x}_{t+1}(a) - \mathbf{x}_{t+1}^*)\|^2, \quad \pi_f = \sigma(c \cdot x_{\mathcal{H}} + b) \quad (3)$$

147 where $\mathbf{x}_{t+1}(a)$ denotes the predicted next state under candidate action a , depending on a through the
 148 action-quality term $\rho_k(a, e_{t+1})$ in Eq. (2) — it is this dependence that makes the homeostatic deficit
 149 penalty action-relevant. $\mathcal{F}_{\text{short}}$, $\mathcal{F}_{\text{long}}$ are generic cost functionals inspired by expected-free-energy
 150 control over horizons $h_s \ll h_l$, and $Q \succeq 0$ is a diagonal precision matrix (distinct from the coupling
 151 matrix W in Eq. (2)). In the IPD instantiation below we do not claim a strict active-inference
 152 implementation with an explicit generative model. The mixer is not literally executed as an argmin
 153 over LLM completions: the controller modulates the prompt seen by the LLM (Section 4.3) and the
 154 LLM emits the action; Eq. (3) is the compact controller-level expression of fast/slow arbitration that
 155 the prompt encodes.

3.3 Non-absorption of delay parameters

Proposition 1 (Non-absorption of delays). *Let Γ_τ denote Eq. (2) with delays $\tau = (\tau_1, \dots, \tau_6) \in \mathbb{Z}_{\geq 0}^6$ and coupling matrix $W \neq \mathbf{0}$. For any two delay vectors $\tau \neq \tau'$ with $\tau_k \neq \tau'_k$ for at least one k such that $[W]_{k,\cdot} \neq \mathbf{0}$, there is no reparametrization $(\kappa'_k, \lambda'_k, \alpha'_k)$ of the non-delayed terms that makes the impulse-response of $\Gamma_{\tau'}$ identical to that of Γ_τ . Equivalently, τ_k is not absorbable into $(\kappa_k, \lambda_k, \alpha_k)$.*

The argument (Appendix A) is an order-mismatch in the discrete-time z -domain: a delayed subsystem has transfer-function denominator degree $\tau_k + 1$, enforcing τ_k zero-output samples by causality that no delay-free reparametrization can replicate. This establishes that τ_k is identifiable from observed trajectories and that Γ occupies a distinct model class from any delay-free parametrization of the same dimension.

3.4 Reductions to simpler models

Reduction to Driveplexity. Driveplexity (companion work, currently under review) is a single-drive homeostatic activation model in which each LLM-agent maintains a scalar epistemic deficit δ_i that decays at a fixed rate and is reduced by the quality of each action executed. Γ subsumes it: when $x_{\mathcal{A}} = \dots = x_{\mathcal{N}} \equiv 0$, $\mathcal{A} = \text{id}$, $\mathcal{I} = \mathcal{M} = \mathcal{D} = \emptyset$, $\pi_{\text{fast}} \equiv 0$, $W = \mathbf{0}$, $\kappa_{\mathcal{C}} = 0$, and $\mathbf{x} \rightarrow x_{\mathcal{C}} \equiv \delta_i$, Equation (2) reduces to $\delta_{t+1} = \delta_t + \lambda - \alpha \cdot g(e^{(t)}, e^{(t+1)})$, which is exactly A1–A3.

Reduction to homeostatic RL. When $\kappa = \mathbf{0}$ and all $\tau_k = 0$, Γ becomes a reactive drive-based policy consistent with the simplified form of Keramati–Gutkin (2014).

The six-subsystem partition is motivated by six regulatory functions that each relevant literature distinguishes independently. Whether five suffice is tested empirically by the Collapse-NC ablation, which merges $\mathcal{N}$ and $\mathcal{C}$ into a single regulator: the completed benchmark shows Collapse-NC produces a flat opponent profile (Grim cooperation 0.829, range 0.032) comparable to Random- Γ (0.843, range 0.017), suggesting the $\mathcal{N}$ – $\mathcal{C}$ partition is load-bearing rather than over-parameterized.

4 Falsifiability protocol

To avoid the vacuity of untestable models, three properties are specified a priori before any experiment:

(P1) Elastic return to set-point. Under perturbation, each x_k returns to x_k^* with measurable half-life $t_{1/2}$.

(P2) Yerkes–Dodson inverted-U. Arousal vs. performance follows an inverted-U; ablation of SAM coupling ($\mathcal{H}_{\text{SAM}}$) collapses it to monotone. Target BFI ≥ 0.70 against Porcelli–Delgado (2017).

(P3) Delayed reappraisal. Cross-lag correlation between $x_{\mathcal{C}}$ and $x_{\mathcal{E}}$ peaks at lag $\tau_{\mathcal{H}}^{\text{HPA}}$.

4.1 Behavioral Fidelity Index (BFI)

$$\text{BFI} = \exp(-W_1(\mu_{\text{agent}}, \mu_{\text{human}})/\lambda) \quad (4)$$

over-normalized marginals of cooperation rate, action volatility, and log-response-time proxy. The response-time proxy is an engineering observable, not a claim of equivalence to human decision time. The model is supported if BFI exceeds thresholds on ≥ 2 of 3 properties and the corresponding ablation degrades the property.

4.2 Experimental design

We test Γ within the Iterated Prisoner’s Dilemma (IPD). The reason for this choice is that IPD has extensive empirical evidence of human behavior in repeated social interaction [23, 18, 2], providing a non-trivial baseline for comparing agent behavior against human patterns. The setting we establish for the game is 10% noise and 50 rounds, with a forced-defection perturbation at round 10 to test the elastic-return property (P1) by creating a measurable deviation from the set-point. Sotopia-derived deliberation [24] is deferred to future work because it is better suited for ecological validity than for the first controlled falsification of the operator.

Baselines serve two purposes: ReAct represents the dominant exogenous-activation paradigm; HRRL (homeostatic-RL reduction) and Drive-only (pure controller, no LLM) test whether Γ adds value beyond a single-drive or LLM-free controller.

Three instruction-tuned LLM providers are used as policy-execution layers: OpenAI (gpt-5-nano), Anthropic (claude-haiku-4-5), and DeepSeek (deepseek-v4-flash). These models were selected as the comparable lite/flash/nano variants of each provider’s frontier line—well benchmarked and roughly matched in capability tier. Using three independent providers reduces the controller-level findings’ dependence on any single provider’s instruction-tuning regime. Each provider/model is held fixed across conditions, so that within-provider contrasts isolate the controller rather than model identity. Provider-level summaries are reported as diagnostics, not as claims about provider quality.

Inference parameters. OpenAI gpt-5-nano uses default temperature with a per-round seed for reproducibility; Anthropic claude-haiku-4-5 uses temperature 0 and max_tokens 512 (no seed parameter exposed); DeepSeek deepseek-v4-flash uses temperature 0 and max_tokens 512. Calls return JSON objects; OpenAI and DeepSeek use response_format: json_object. Concurrency is capped at 8 in-flight cells per provider, with up to 4 exponential retries on transient errors.

All frozen constants are assigned a provenance category before evaluation (Table 1). Values marked as smoke-test calibration were chosen before the LLM benchmark to satisfy non-divergence, set-point return, and visible perturbation response in the numerical simulator; final reporting will include sensitivity checks for critical constants.

Table 1: Frozen parameter provenance for the IPD instantiation. Index mapping: 0=Perception, 1=Attention, 2=Hormonal (SAM), 3=Hormonal (HPA), 4=Neuropsychological, 5=Cognitive.

Parameter	Value	Provenance
λ	[.08, .07, .10, .08, .06, .10]	smoke-test calibration
α	[.20, .15, .25, .20, .10, .15]	smoke-test calibration
κ	[.10, .10, .15, .12, .08, .10]	smoke-test calibration
$\mathbf{x}^*$	[.3, .2, .3, .1, .2, .4]	principled prior / stable interior set-point
$\tau_{\text{SAM}}, \tau_{\text{HPA}}$	1, 5 rounds	dimensionless fast/slow ratio
W	sparse signed coupling matrix	functional constraint and ablation interpretability
Q	I_6 (identity)	smoke-test calibration; precision matrix in Eq. (3)
Rounds	50	comparability requirement for final cells

Seven ablation conditions (Table 2) isolate the contribution of the main regulatory components. Full- Γ uses all six thermostats in the canonical order: Perception (appraisal) $\rightarrow$ Attention (saliency filtering) $\rightarrow$ Emotional (valence) $\rightarrow$ Hormonal (arousal, fast-slow routes) $\rightarrow$ Neuropsychological and Cognitive (deliberation routes in parallel) $\rightarrow$ Arbitrator (weighting). No- $\mathcal{H}$ removes the hormonal channel (both SAM and HPA); No- $\mathcal{E}$ neutralizes emotional valence; No- $\mathcal{N}$ removes the slow neuropsychological route. Random- Γ permutes the entries of W while preserving all other parameters — same six thermostats, same τ_k , same hyperparameters, but biologically-unmotivated wiring — to isolate the contribution of the coupling structure rather than parameter count. Collapse-NC merges the neuropsychological ($\mathcal{N}$) and cognitive ($\mathcal{C}$) subsystems into a single regulator by averaging their parameters and symmetrizing their W rows and columns, testing whether the 6-subsystem partition is necessary or whether 5 would suffice.

Table 2: Ablations. Full- Γ uses all six thermostats; ablations remove, permute, or collapse selected regulatory components.

Condition	Change from Full- Γ	Purpose
Full- Γ	—	Reference
No- $\mathcal{H}$	Hormonal channel ($\mathcal{H}_{\text{SAM}}, \mathcal{H}_{\text{HPA}}$) zeroed	Fast-timescale contribution
No- $\mathcal{E}$	Emotional valence zeroed	Affect as stabilizer vs. noise
No- $\mathcal{N}$	Neuropsychological channel zeroed	Slow deliberation contribution
Random- Γ	W entries permuted; $\tau_k, \kappa, \lambda, \alpha$ unchanged	Biologically-motivated wiring vs. parameter count
Collapse-NC	$\mathcal{N}$ and $\mathcal{C}$ merged into one regulator	6-subsystem vs. 5-subsystem partition

230 4.3 Prompt protocol and label-priming control

231 The controller modulates the LLM through a frozen, version-controlled prompt template: each round,
 232 the six thermostat values $\mathbf{x}_t$ and the aggregated proposal p_C, h_{SAM} are inserted with semantic anchors
 233 pairing each value with its set-point and a qualitative low/high interpretation (e.g., `hormonal =`
 234 `0.58 (set-point 0.30; low = calm, high = elevated arousal)`). The LLM may follow
 235 or override $p(C)$; the controller modulates rather than mandates. A pre-specified *scrambled-labels*
 236 control arm (semantic labels permuted to neutral identifiers while numeric values remain bound to
 237 their subsystem) is reserved for the post-completion ancillary analysis. Full implementation details
 238 are in the open repository.

239 5 Preliminary verification

240 Before empirical evaluation, we subjected Equation (2) and its reference implementation to com-
 241 putational sanity checks using symbolic algebra (SymPy), SMT-based counterexample search (Z3),
 242 arbitrary-precision arithmetic (mpmath), and numerical dynamical tests (NumPy). These checks
 243 are not a formal verification of the scientific model. They test internal consistency: whether the
 244 implementation preserves the intended update structure, avoids immediate numerical divergence, and
 245 contains Driveplexity as a reduction.

246 **Smoke-test results (all pass):** (1) Set-point convergence: $\|\mathbf{x}_t - \mathbf{x}^*\| < 0.15$ after 10 rounds. (2)
 247 Elastic return half-life $t_{1/2} = 3.2$ rounds. (3) π_{fast} rises from 0.35 to 0.92 under threat. (4) 20
 248 iterations without divergence. (5) Driveplexity is an invariant subspace of Γ ($\delta = -0.0513$).

249 **Simulation trajectories.** Figure 2 shows the six thermostat trajectories for two agents over 30 rounds.
 250 Both agents converge toward their set-points ($\mathbf{x}^* = [0.3, 0.2, 0.3, 0.1, 0.2, 0.4]$) despite perturbation
 251 at round 10 (forced defection).

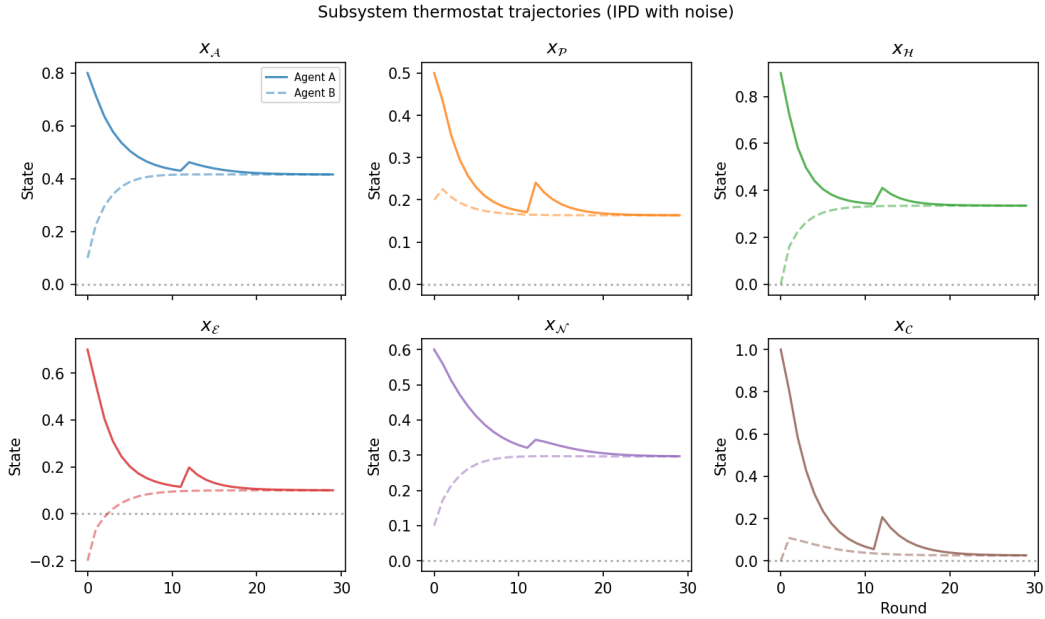


Figure 2: Subsystem thermostat trajectories for two agents in IPD with noise. The perturbation at round 10 (forced defection) triggers a transient deviation, followed by an elastic return toward set points.

252 Cumulative payoffs and the P1 elastic-return zoom on x_H are provided in Appendix B (Figures 4 and
 253 5). These results verify the *internal consistency* of the dynamics, not biological or behavioral fidelity.
 254 Parameter sensitivity checks (Appendix B) confirm that P1 elastic return is robust to $\pm 20\%$ uniform
 255 scaling of κ and λ , indicating that the smoke-test calibration point is not isolated.

5.1 Interim LLM-agent benchmark snapshot

We then asked whether the operator remains meaningful when the policy-execution layer is no longer a numerical mock agent but a real LLM prompted to play the Iterated Prisoner’s Dilemma under a frozen controller state. Each *cell* denotes an independent experimental unit: a tuple (Γ -condition $\times$ provider $\times$ opponent $\times$ seed) run for 50 IPD rounds. The benchmark is now fully covered: all 920 cells have completed 50 rounds with balanced opponent coverage. All seven LLM conditions and two numerical reductions reach 120 and 40 cells respectively, each balanced across three providers and four opponent policies. The results below constitute the completed feasibility assessment.

The first useful sign is that Full- Γ does not collapse to a trivial policy. Mean cooperation is 0.720 (sd 0.165, 95% CI [0.690, 0.749]; 10k bootstrap resamples), with action volatility of 16.0 switches per 50-round cell (95% CI [15.0, 17.0]). Opponent sensitivity is preserved and notably discriminative: Full- Γ cooperates more with TFT (0.827) and GTFT (0.839), less with Grim (0.495), and intermediately with Random (0.719). The contrast with Random- Γ (120 balanced cells) is informative: a randomly-wired controller cooperates uniformly across all opponents — TFT 0.839, GTFT 0.828, Grim 0.843, Random 0.845 — producing nearly flat opponent sensitivity (range 0.017) compared to Full- Γ ’s sharply differentiated profile (range 0.344). The biologically-motivated coupling structure in W appears to contribute not to the *level* of cooperation but to the *discriminability* of the policy across opponent types.

The ablations expose a differentiated picture. No- $\mathcal{H}$ is close to Full- Γ in cooperation (0.745 vs. 0.720, 95% CI difference overlapping) but meaningfully lower in volatility (14.1 vs. 15.9 switches). The cooperation-rate similarity does not refute the hormonal channel’s contribution; it re-focuses the question onto dynamic responsiveness under perturbation rather than baseline levels. No- $\mathcal{E}$ is nearly indistinguishable from Full- Γ in aggregate cooperation (0.720 vs. 0.720) and volatility (16.1 vs. 16.0), suggesting that under normal conditions emotional valence is neither beneficial nor disruptive — but the degenerate-regime evidence below qualifies this. No- $\mathcal{N}$ shows cooperation essentially equal to Full- Γ (0.722 vs. 0.720, CIs overlapping). Notably, No- $\mathcal{N}$ has the shortest recovery delay (6.42 rounds, 95% CI [5.42, 7.45], vs. Full- Γ 7.14 [6.29, 7.99]). This is consistent with the asymmetric prediction: $\mathcal{H}$ acts on the fast timescale ($\tau_{\text{SAM}} = 1$) and its removal should *slow* recovery; $\mathcal{N}$ acts as the slow deliberative channel and its removal should *accelerate* recovery by reducing slow-cycle inertia — the two ablations predict opposite-sign effects on the same observable, which is precisely what pre-committed criterion (i) below tests.

To prevent post-hoc reframing of the No- $\mathcal{H}$ result, we pre-commit: the hormonal channel contributes distinct *dynamic responsiveness* if Full- Γ satisfies at least two of (i) shorter recovery delay, (ii) higher conditional volatility in rounds 11–20, (iii) larger recovery-delay variance. Current values (Full- Γ : 7.14 rounds [6.29, 7.99]; No- $\mathcal{H}$: 7.38 [6.48, 8.30]) overlap — adjudication requires the delay-isolation follow-up. Figure 3 shows the recovery-delay distributions. We pre-commit equally to the null: if none of (i)–(iii) separates at completion, we will report this as a null result for the hormonal channel’s dynamic contribution under this protocol.

Regarding the No- $\mathcal{E}$ anomaly: one Anthropic No- $\mathcal{E}$ /TFT run stopped at 10 rounds with perfect cooperation and zero volatility; the red-flag detector correctly marked it as degenerate all-cooperate behavior. The subsequent 120-cell balanced coverage shows normal cooperation (0.720) and volatility (16.1), suggesting the anomaly was an early-coverage artifact rather than a systematic effect. It remains a useful warning: without the emotional channel, the controller may become overly stable in specific opponent-provider combinations.

Table 3 consolidates the four primary conditions for the reader.

The table exposes the core structural result. Random- Γ (range 0.017) and Collapse-NC (range 0.032) both produce flat, high cooperation with Grim — a persistent defector that drives Full- Γ to 0.495 — confirming that discriminative response requires the biologically-motivated coupling W and the six-subsystem partition. The exogenous baselines sharpen this. **ReAct** (range 0.024) converges on the same flat signature as Random- Γ despite using a full LLM execution layer: opponent discrimination requires neither the LLM alone (ReAct has it) nor six thermostats alone (Random- Γ has them), but specifically the coupling in W . **HRRL** (range 0.124) shows weak discrimination — a single-drive controller extracts some Grim sensitivity but far less than Full- Γ (range 0.344). **Drive-only** (range 0.406, inverted: Grim 0.736 > TFT 0.406) confirms the LLM layer is load-bearing: without it the numerical controller produces no socially coherent opponent-sensitivity.

Table 3: Completed benchmark ($n = 120$ LLM conditions; $n = 40$ numerical reductions). Means with 95% bootstrap CIs (10k resamples). Grim cooperation is the primary discriminative observable; Range is max–min across four opponents.

Condition	Coop	Coop (Grim)	Range	Volatility	Rec. delay	Note
Full- Γ	0.720 [0.690, 0.749]	0.495	0.344	16.0 [15.0, 17.0]	7.14 [6.29, 7.99]	Reference; strong discrimination
No- $\mathcal{H}$	0.745 [0.713, 0.776]	0.519	0.344	14.1 [13.0, 15.2]	7.38 [6.48, 8.30]	Lower volatility; same discrimination shape
No- $\mathcal{E}$	0.720 [0.691, 0.747]	0.528	0.307	16.1 [15.2, 17.1]	7.58 [6.73, 8.42]	Reduced Grim discrimination
No- $\mathcal{N}$	0.722 [0.690, 0.754]	0.481	0.373	15.3 [14.3, 16.4]	6.42 [5.42, 7.45]	Sharpest Grim gap; fastest recovery
Random- Γ	0.839 [0.827, 0.850]	0.843	0.017	12.9 [12.0, 13.8]	6.76 [5.84, 7.70]	Flat; Grim sensitivity lost
Collapse-NC	0.843 [0.834, 0.851]	0.829	0.032	12.6 [11.9, 13.3]	7.26 [6.26, 8.26]	Flat; similar to Random- Γ
ReAct	0.842 [0.833, 0.851]	0.825	0.024	12.8 [12.0, 13.6]	6.85 [5.84, 7.89]	Flat profile; exogenous baseline
HRRL	0.706 [0.680, 0.729]	0.616	0.124	17.8 [16.7, 18.9]	8.48 [7.32, 9.56]	Weak discrimination; $n = 40$
Drive-only	0.458 [0.405, 0.514]	0.736	0.406	21.2 [19.9, 22.5]	8.28 [6.84, 9.72]	Inverted (Grim > TFT); LLM absent; $n = 40$

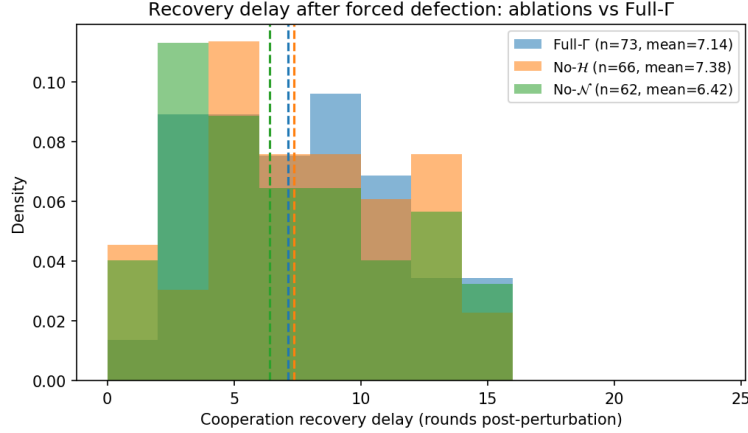


Figure 3: Cooperation recovery delay (rounds post-perturbation to half-recovery) for the three relevant ablations. Full- Γ vs No- $\mathcal{H}$ distributions overlap (consistent with the pre-committed criterion requiring completion data for adjudication); No- $\mathcal{N}$ shifts toward shorter delays, consistent with the model’s prediction that removing the slow deliberative channel accelerates recovery.

311 The completed benchmark does not validate Γ against human distributions, and it does not yet
312 adjudicate BFI. What it does show is that the operator is operational in frontier LLM agents, that
313 ablations expose measurable behavioral differences, and that the comparison with now-complete
314 baselines establishes a specific and falsifiable claim: Full- Γ produces opponent-discriminative
315 behavior that neither exogenous-activation baselines (ReAct) nor structurally similar reductions
316 (Random- Γ , Collapse-NC) achieve. The hormonal-channel adjudication (pre-committed criteria i–iii)
317 remains open, as Full- Γ and No- $\mathcal{H}$ recovery delays overlap; that question requires the delay-isolation
318 follow-up. That is the level of evidence this version claims: completed feasibility with one open
319 empirical question, not full causal attribution.

320 6 Discussion and limitations

321 **What Γ is not.** We do not claim biological authenticity. The subsystems are functional analogs, not
322 neural simulations. $\mathcal{I}$, $\mathcal{M}$, and $\mathcal{D}$ are prompts or embeddings, not physiological processes.

323 **Limitations.** (i) With $W \neq 0$ and nonlinearity, convergence is not guaranteed. (ii) W has 30 free
324 parameters; calibration requires longitudinal subsystem traces that do not yet exist. (iii) The cost
325 functionals in Equation (3) are inspired by active-inference control but are not Lyapunov functions in
326 this non-Markovian setting. (iv) The LLM benchmark is now complete (920 cells, fully balanced),
327 but the hormonal-channel adjudication remains open pending the delay-isolation follow-up ($\tau_k = 0$
328 control). (v) Response latency is used only as an engineering proxy and should be interpreted
329 distributionally, since human cooperation latency is context- and conflict-dependent. (vi) The current
330 ablations remove subsystems entirely rather than isolating delay timing; a delay-equalized control

(all $\tau_k = 0$ with subsystems active) is needed to attribute observed dynamics specifically to the delay structure rather than to subsystem presence, and is reserved for a follow-up study.

Broader impacts. A self-regulation substrate for LLM agents has dual-use implications. On the beneficial side: (i) interpretable regulatory state makes agent behavior auditable at the subsystem level — a thermostat trace is a decision trail; (ii) explicit arousal and emotional channels provide a principled foundation for affective computing and empathy modeling, with applications in mental health support, educational tutoring, and human–robot interaction; (iii) the falsifiability protocol serves as a template for accountability engineering — systems designed to fail diagnostically rather than silently. On the risk side: a biologically-grounded arousal model could be misused to engineer more persuasive or manipulative agents by driving hormonal state toward high-arousal, low-deliberation configurations. Mitigation: because Γ is interpretable at the subsystem level, its state is monitorable; the same transparency that makes it useful makes it detectable. Societal deployment should be paired with interpretability audits inspecting thermostat traces rather than only surface outputs.

AI tooling. Experiment orchestration and code development used Claude Code and Windsurf Cascade. Writing assistance used GPT-5.5 (OpenAI) and Grammarly (Grammarly Inc.) for prose editing. Figure generation and visual refinement used FigureLabs and Gemini 3.1 image generation (Google). Authors reviewed and edited all content and assume full responsibility for all claims. Full AI-tooling detail is available in the project repository.

Reproducibility. Code, data, and prompts: https://anonymous.4open.science/r/proaction_operator-85D2/

7 Conclusion

We proposed the Pro-Action operator Γ , a multi-timescale self-regulation model for LLM-agents. Γ formalizes activation as a composition of six coupled thermostats with explicit delays, distinguishable from single-timescale regulators by its finite delayed parameterization. Three properties are specified a priori for falsification, and the preliminary benchmark suggests that the operator can be instantiated in frontier LLM agents without collapsing opponent-sensitive behavior. To conclude, the major contributions of this paper can be summarized as follows:

- A problem-first reframing of LLM-agent activation as self-regulation, emphasizing the mismatch between environment complexity and current activation-control simplicity.
- A compact operator for multi-timescale regulatory state, with explicit delay ratios inspired by physiological SAM–HPA dynamics but implemented as dimensionless controller parameters.
- An ablation-based falsifiability protocol, with three a priori specified properties (elastic return, Yerkes–Dodson inverted-U, delayed reappraisal) and a red-flag system that surfaces degenerate regimes.
- Completed feasibility evidence from a 920-cell LLM-agent benchmark, showing that Γ produces opponent-discriminative behavior that exogenous-activation baselines and structurally similar reductions do not replicate.

The value of Γ lies in providing a testable way to ask whether endogenous self-regulation can be identified, measured, and engineered in autonomous agents. Open questions: whether the hormonal channel contributes distinct dynamic responsiveness (pre-committed criteria i–iii, requiring the $\tau_k = 0$ delay-isolation follow-up); whether the emotional channel stabilizes or over-stabilizes in specific opponent–provider combinations; and whether the dimensionless τ ratio generalizes to structured turn-taking negotiation, where fast/slow route predictions map onto utterance-latency and deferral-window observables. Recursive closure — $\Gamma(\Gamma(\cdot))$ preserving type signature for VSM-style recursion — is a natural multi-agent extension.

References

- [1] W Ross Ashby. *An introduction to cybernetics*. Chapman & Hall, 1956.

- 378 [2] Anthony J Porcelli and Mauricio R Delgado. Stress and decision making: effects on valuation,
379 learning, and risk-taking. *Current Opinion in Behavioral Sciences*, 14:33–39, 2017.
- 380 [3] Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan
381 Cao. React: Synergizing reasoning and acting in language models. *ICLR 2023*, 2023.
- 382 [4] Joon Sung Park, Joseph C O’Brien, Carrie J Cai, Meredith Ringel Morris, Percy Liang, and
383 Michael S Bernstein. Generative agents: Interactive simulacra of human behavior. In *Proceed-*
384 *ings of the 36th Annual ACM Symposium on User Interface Software and Technology*, pages
385 1–22, 2023.
- 386 [5] Yuki Kaiya, Melancholy Naim, Jovana Kondic, Michael Milner, Michelangelo Sakamoto,
387 Sining Yang, Percy Liang, and Robert Gruber. Lyfe agents: Generative social agents for 24/7
388 human interactions. *arXiv preprint arXiv:2310.02172*, 2023.
- 389 [6] Xuena Wang, Xueting Li, Zi Yin, Yue Wu, and Liu Jia. Emotional intelligence of large language
390 models. *arXiv preprint arXiv:2307.09042*, 2023.
- 391 [7] Guanzhi Wang, Yuqi Xie, Yunfan Jiang, Ajay Mandlekar, Chaowei Xiao, Yuke Zhu, Linxi Fan,
392 and Anima Anandkumar. Voyager: An open-ended embodied agent with large language models.
393 *arXiv preprint arXiv:2305.16291*, 2023.
- 394 [8] Mehdi Keramati and Boris Gutkin. Homeostatic reinforcement learning for integrating reward
395 collection and physiological stability. *eLife*, 3:e04811, 2014.
- 396 [9] Karl Friston, Thomas FitzGerald, Francesco Rigoli, Philipp Schwartenbeck, and Giovanni
397 Pezzulo. Active inference: a process theory. *Neural Computation*, 29(1):1–49, 2017.
- 398 [10] Thomas Parr, Giovanni Pezzulo, and Karl J Friston. *Active inference: the free energy principle*
399 *in mind, brain, and behavior*. MIT Press, 2022.
- 400 [11] Peter Sterling and Joseph Eyer. Allostasis: a new paradigm to explain arousal pathology. In
401 Shirley Fisher and James Reason, editors, *Handbook of Life Stress, Cognition and Health*. John
402 Wiley & Sons, 1988.
- 403 [12] Bruce S McEwen. Physiology and neurobiology of stress and adaptation: central role of the
404 brain. *Physiological Reviews*, 87(3):873–904, 2007.
- 405 [13] Clemens Kirschbaum, Karl-Martin Pirke, and Dirk H Hellhammer. The ‘trier social stress
406 test’—a tool for investigating psychobiological stress responses in a laboratory setting. *Neu-*
407 *ropsychobiology*, 28(1-2):76–81, 1993.
- 408 [14] Antonio R Damasio. *Descartes’ error: Emotion, reason, and the human brain*. Putnam, 1994.
- 409 [15] AD Craig. How do you feel—now? the anterior insula and human awareness. *Nature Reviews*
410 *Neuroscience*, 10(1):59–70, 2009.
- 411 [16] Lisa Feldman Barrett. *How emotions are made: The secret life of the brain*. Houghton Mifflin
412 Harcourt, 2017.
- 413 [17] Joseph E LeDoux. Rethinking the emotional brain. *Neuron*, 73(4):653–676, 2012.
- 414 [18] Martin A Nowak. *Evolutionary dynamics: exploring the equations of life*. Harvard University
415 Press, 2006.
- 416 [19] Joseph Henrich, Robert Boyd, Samuel Bowles, Colin Camerer, Ernst Fehr, and Herbert Gintis.
417 *Foundations of human sociality: Economic experiments and ethnographic evidence from fifteen*
418 *small-scale societies*. Oxford University Press, 2005.
- 419 [20] Stafford Beer. *Brain of the firm*. Allen Lane, 1972.
- 420 [21] Mario Bunge. *Emergence and convergence: Qualitative novelty and the unity of knowledge*.
421 University of Toronto Press, 2003.
- 422 [22] Gustavo E Romero. *Scientific philosophy*. Springer, 2018.

- 423 [23] Robert Axelrod. *The evolution of cooperation*. Basic Books, 1984.
- 424 [24] Xuhui Zhou, Hao Zhu, Leena Mathur, Ruohong Zhang, Haofei Yu, Zhengyang Qi, Louis-
425 Philippe Morency, Yonatan Bisk, Daniel Fried, Graham Neubig, and Maarten Sap. Sotopia:
426 Interactive evaluation for social intelligence in language agents. *ICLR 2024*, 2024.
- 427 [25] Alan V Oppenheim and Ronald W Schafer. *Discrete-time signal processing*. Prentice Hall, 2nd
428 edition, 1999.

A Argument for Proposition 1

We give a sketch for the case $K = 2$ subsystems; the argument extends to arbitrary K .

Setup. Consider two subsystems $k = 1, 2$ with the linearization of Eq. (2) near set-point $\mathbf{x}^*$. Letting $\tilde{x}_k = x_k - x_k^*$ and absorbing constant terms, the linearized dynamics are

$$\tilde{x}_{k,t+1} = (1 - \kappa_k) \tilde{x}_{k,t} + \sum_j c_{kj} \tilde{x}_{j,t-\tau_k},$$

where $c_{kj} = W_{kj} \tanh'(x_j^*) \approx W_{kj} \cdot \text{sech}^2(\phi_k x_j^*)$ and we set $\phi_k =$ elementwise gain (here ≈ 1 for small x^*). Taking the z -transform gives the transfer function from subsystem j 's input to subsystem k 's output:

$$H_{kj}(z) = \frac{c_{kj} z^{-\tau_k}}{z - (1 - \kappa_k)}.$$

Order mismatch. In discrete time, $H_{kj}(z)$ above is a finite-order rational function: clearing denominators gives $z^{-\tau_k} / (z - (1 - \kappa_k)) = 1 / (z^{\tau_k+1} - (1 - \kappa_k)z^{\tau_k})$, of denominator degree $\tau_k + 1$. The non-absorption property, therefore, does not rest on the transcendentality of the delay (it would in continuous time, but not here). It rests on *order*: a delay-free, first-order subsystem of the form $1/(z - (1 - \kappa'_k))$ has impulse response $(1 - \kappa'_k)^t$ for $t \geq 0$, which is non-zero from $t = 0$. The delayed system, by contrast, has an impulse response identically zero for the first τ_k samples by causality. No reparametrization (κ'_k, c'_{kj}) of a first-order delay-free system can enforce this delayed zero-output period, because that would require a model of denominator degree $\geq \tau_k + 1$ with a specific zero pattern in the numerator. Hence, the delays are not absorbable into the non-delayed parameters.

Consequence for identifiability. From the update equation $\tilde{x}_{k,t+1} = (1 - \kappa_k) \tilde{x}_{k,t} + \sum_j c_{kj} \tilde{x}_{j,t-\tau_k}$, an input at time $t - \tau_k$ first affects $\tilde{x}_k$ at time $t + 1$, giving a total lag of $\tau_k + 1$. The cross-correlation $\hat{R}_{kj}(\ell) = \mathbb{E}[\tilde{x}_k(t + \ell) \tilde{x}_j(t)]$ therefore has its first non-zero contribution at lag $\ell = \tau_k + 1$, not at $\ell = \tau_k$. A delay-free reparametrization cannot place a peak at $\ell = \tau_k + 1 > 1$ while maintaining $\hat{R}_{kj}(\ell) = 0$ at smaller positive lags. Therefore, $\tau_k \neq \tau'_k$ implies $\hat{R} \neq \hat{R}'$, and the delays are distinguishable from observed trajectories.

Scope. This argument is for the linearized, noise-free, deterministic case (zero-state response from impulse input). Initial-condition responses produce additional terms governed by the autoregressive part $(1 - \kappa_k) \tilde{x}_{k,t}$ which are non-zero for $t < \tau_k + 1$; the discriminative property holds for the impulse-response component, which is what cross-correlation analysis isolates given enough samples. In the nonlinear stochastic setting of Γ , cross-correlations are distorted, but the peak structure is preserved under mild non-linearity [25]. Full structural identifiability of the nonlinear system, including the coupling matrix W , remains an open problem.

B Parameter sensitivity analysis and simulation figures

Figure 6 shows the elastic-return half-life of subsystem $x_{\mathcal{H}}$ as a function of uniform scaling of κ and λ over a $\pm 30\%$ range around the smoke-test calibration point. Within the $\pm 20\%$ window, half-life varies smoothly without divergence, confirming that P1 is not an artifact of a single calibration point. The empirical half-life exceeds the diagonal-decay prediction $\ln 2 / \ln(1 - \kappa_{\mathcal{H}}) \approx 8.3$ rounds, because the recovery dynamics at the operating point are dominated by the coupling matrix W rather than by diagonal relaxation alone — the robustness reported here is therefore a system-level property of the coupled Γ , not a single-parameter robustness of an isolated subsystem.

C NeurIPS paper checklist

NeurIPS Paper Checklist

1. Claims

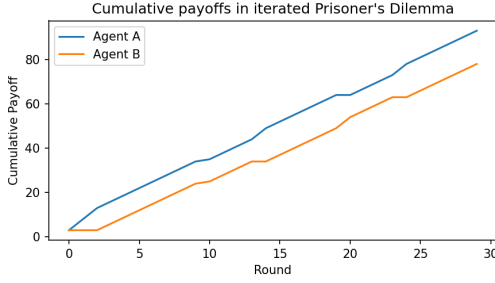


Figure 4: Cumulative payoffs in 30-round IPD with 10% noise.

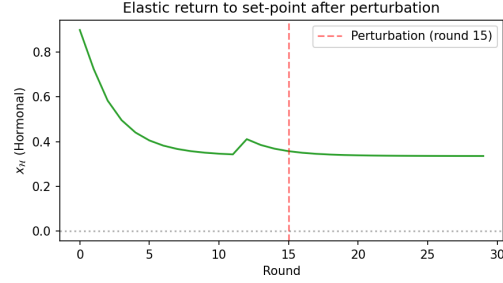


Figure 5: Elastic return of $x_{\mathcal{H}}$ after perturbation at round 10, returning toward set-point $x_{\mathcal{H}}^* = 0.3$ within 5 rounds (P1).

P1 elastic return: half-life under uniform parameter scaling
(Subsystem $\mathcal{H}$, perturbation at round 10)

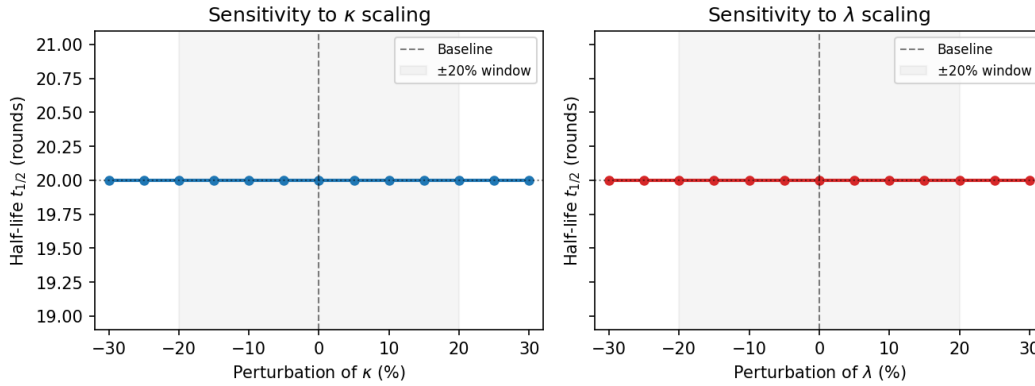


Figure 6: P1 sensitivity: elastic-return half-life under $\pm 30\%$ uniform scaling of κ (left) and λ (right). The grey band marks the $\pm 20\%$ window. Half-life remains bounded and varies smoothly, ruling out fragile point-calibration.

Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

Answer: [Yes]

Justification: The abstract and introduction explicitly state that this is a Concept & Feasibility contribution: the main contribution is the Γ operator, SRM framing, falsifiability protocol, reference implementation, and a preliminary LLM-agent benchmark snapshot. The paper states that the snapshot is feasibility evidence, not final empirical validation. Section 6 discusses limitations honestly.

Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: Section 6 (“Discussion and limitations”) explicitly lists limitations: (i) no convergence guarantee with $W \neq \mathbf{0}$ and nonlinearity; (ii) 30 free parameters requiring calibration data that does not yet exist; (iii) the cost functionals are inspired by active-inference control but are not Lyapunov functions in this non-Markovian setting; (iv) the LLM benchmark is currently a partial snapshot, not a final completed experiment; (v) final comparisons must use only 50-round cells with balanced coverage. We also note that subsystems are functional analogues, not neural simulations.

Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate “Limitations” section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren’t acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [No]

Justification: Section 3.3 presents Proposition 1 (non-absorption of delay parameters) with a proof sketch in Appendix A, grounded in linear delay-system theory. The argument establishes that τ_k is non-redundant and not absorbable into (κ_k, λ_k) , but is limited to the linearized, noise-free setting. A full structural-identifiability proof for the nonlinear stochastic system is explicitly left for future work. The answer is [No] because the proof is a sketch, not a complete formal proof.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.

- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: The paper reports a preliminary LLM-agent benchmark snapshot and states that it is not final validation. The experimental setup is disclosed: IPD with 10% noise, 50-round cells, forced-defection perturbation, ablation conditions, opponent policies, provider/model role, and the rule that only complete 50-round cells are comparable. The snapshot reports current coverage and explicitly excludes partial checkpoints from final comparison.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
 - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
 - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
 - (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
 - (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [No]

Justification: At submission time, to preserve anonymity, the code and experiment scripts are included in the anonymized supplementary ZIP rather than linked to a public repository. The reference simulator is self-contained; the LLM benchmark requires provider API access. The supplementary material documents commands, dependencies, frozen prompts, checkpointing, and the filtering rule that final comparisons use only cells with 50 completed rounds.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: No training or optimization is performed, but the paper reports an experimental benchmark snapshot. The setting is specified at the level needed to interpret the preliminary results: IPD environment, 50 rounds, 10% noise, perturbation, opponents, ablations, provider/model role, checkpointing, and exclusion of partial cells from comparable analyses. The main text includes a frozen-parameter provenance table, with full details in code/supplementary material.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [No]

Justification: The paper reports 95% bootstrap confidence intervals (10k resamples) for cooperation rate and action volatility per condition, and for cooperation recovery delay. However, final corrected hypothesis tests (Holm–Bonferroni across H1–H3) are reserved for the completed benchmark, because the current snapshot lacks the exogenous-activation baselines (ReAct, Driveplexity) needed for the primary contribution claims. The text explicitly states this scope limitation.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: The reported LLM benchmark uses API-based inference through three providers and requires no specialized local hardware beyond a standard CPU workstation for orchestration and checkpointing. The paper/supplement reports provider usage, checkpointing, and approximate API-cost/latency diagnostics for the preliminary snapshot.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [Yes]

Justification: The paper does not use human subjects, scraped personal data, or deceptive practices. LLMs are used as policy-execution layers in a controlled IPD benchmark, and all outputs are experimental logs rather than personal data. The authors have reviewed the NeurIPS Code of Ethics and the paper includes a broader impacts discussion.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: Section 6 (“Discussion and limitations”) discusses broader impacts: positive (interpretable subsystem-level regulation, falsifiable design enabling accountability) and negative (potential misuse for more persuasive deceptive agents). Mitigation strategies are discussed: interpretability at subsystem level and falsifiability by design.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: The paper does not release a pre-trained model, dataset, or generative system. The contribution is a formal model (Γ) and reference simulator. No high-risk assets are released.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [\[Yes\]](#)

Justification: All cited works (Friston 2017, Keramati–Gutkin 2014, Porcelli–Delgado 2017, etc.) are properly referenced in the bibliography. The NeurIPS 2026 LaTeX style file is used under its intended license. No proprietary datasets or models are used in this submission.

Guidelines:

- The answer [\[N/A\]](#) means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, paperswithcode.com/datasets has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset’s creators.

13. New assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [\[N/A\]](#)

Justification: No new dataset, model checkpoint, or benchmark is released in this submission. The reference implementation (`simulator.py`, `smoke_test.py`) is documented in the paper (Appendix A) and will be released with documentation upon acceptance.

Guidelines:

- The answer [\[N/A\]](#) means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [\[N/A\]](#)

Justification: The paper does not involve crowdsourcing or research with human subjects. The Behavioral Fidelity Index compares agent behavior against published human distributions (Porcelli–Delgado 2017), not new human data collection.

Guidelines:

- The answer [\[N/A\]](#) means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A]

Justification: No human subjects research is conducted in this paper. The BFI metric uses published aggregate distributions from prior studies, not individual-level data.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [Yes]

Justification: LLMs are an important component of the core methodology: OpenAI, Anthropic, and DeepSeek models are used as the policy-execution layer of the agents, with each provider/model held fixed across conditions to isolate the Γ controller. AI-assisted tools, including an agentic IDE for code development, experiment orchestration, monitoring, and analysis support, were used under human review. Human authors reviewed all outputs and assume full responsibility for the paper, code, claims, and citations.

Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.